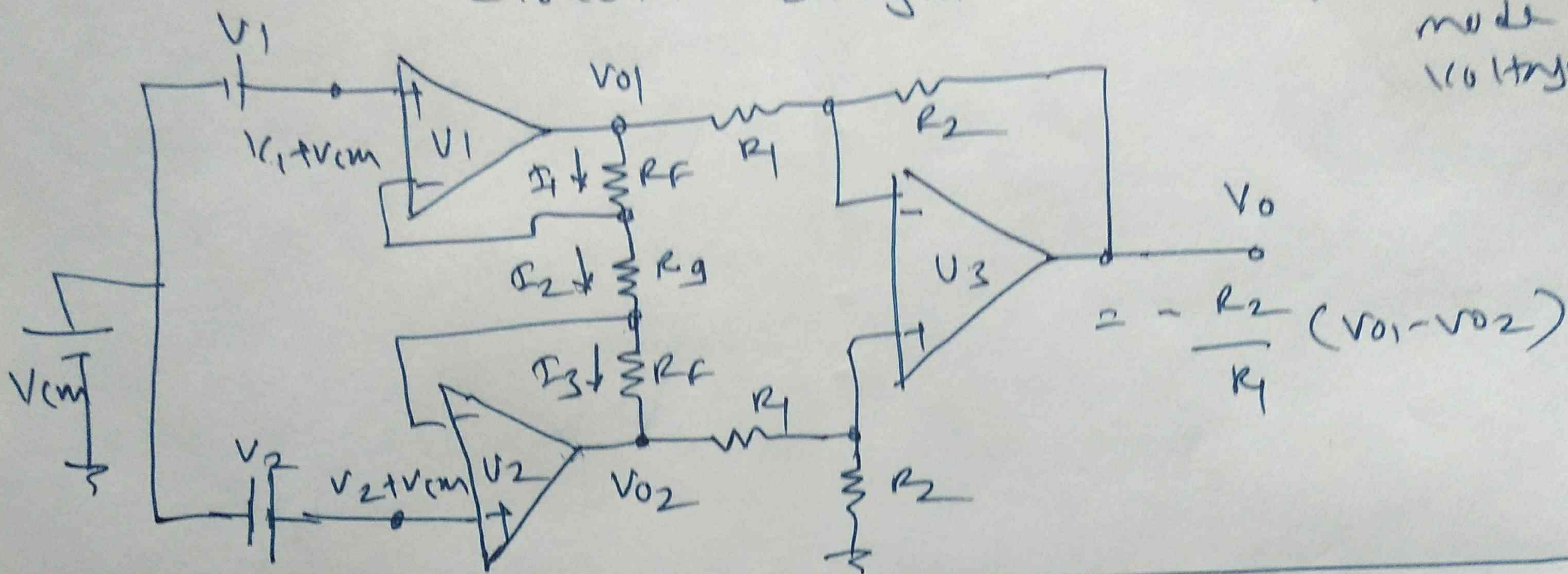


①

Deriving gain equation for 3-OPAMP Instru-Amplifier

circuit diagram

 V_{cm} : common mode voltageWe know $I_1 = I_2 = I_3$ similarly find I_3

$$I_1 = \frac{V_{01} - V_1 - V_{cm}}{R_F}$$

$$I_3 = \frac{V_2 + V_{cm} - V_{02}}{R_F}$$

$$I_2 = \frac{V_1 + V_{cm} - V_2 - V_{cm}}{R_g} = \frac{V_1}{R_g} - \frac{V_2}{R_g}$$

Now compute V_{01}

$$I_1 = I_2$$

$$V_{01} - V_1 - V_{cm} = \frac{R_F}{R_g} V_1 - \frac{R_F}{R_g} V_2$$

& compute V_{02}

$$I_2 = I_3$$

$$\frac{V_1}{R_g} - \frac{V_2}{R_g} = \frac{V_2 + V_{cm} - V_{02}}{R_F}$$

$$V_{01} = V_1 + V_{cm} + \frac{R_F}{R_g} V_1 - \frac{R_F}{R_g} V_2 \quad \rightarrow (1)$$

$$-V_{02} = \frac{R_F}{R_g} V_1 - \frac{R_F}{R_g} V_2 - \frac{V_2 + V_{cm}}{R_F} \times R_F$$

Now App eqn (1) & (2)

$$-V_{02} = \frac{R_F}{R_g} V_1 - \frac{R_F}{R_g} V_2 - V_2 - V_{cm} \rightarrow (2)$$

$$V_{01} - V_{02} = V_1 + V_{cm} + \frac{R_F}{R_g} V_1 - \frac{R_F}{R_g} V_2$$

$$+ \frac{R_F}{R_g} V_1 - \frac{R_F}{R_g} V_2 - V_2 - V_{cm}$$

$$V_{01} - V_{02} = V_1 + \frac{R_F}{R_g} V_1 + \frac{R_F}{R_g} V_1$$

$$- V_2 - \frac{R_F}{R_g} V_2 - V_2 - \frac{R_F}{R_g}$$

$$V_{01} - V_{02} = V_1 \left(1 + \frac{2R_F}{R_g}\right) - V_2 \left(1 + \frac{2R_F}{R_g}\right)$$

$$V_{01} - V_{02} = (V_1 - V_2) \left(1 + \frac{2R_F}{R_g}\right)$$

The o/p of instrumentation Amplifier is $V_0 = -\frac{R_2}{R_1} (V_{01} - V_{02})$

$$\therefore V_0 = -\frac{R_2}{R_1} \times \left(1 + \frac{2R_F}{R_g}\right) (V_1 - V_2)$$

②

Some Discussion

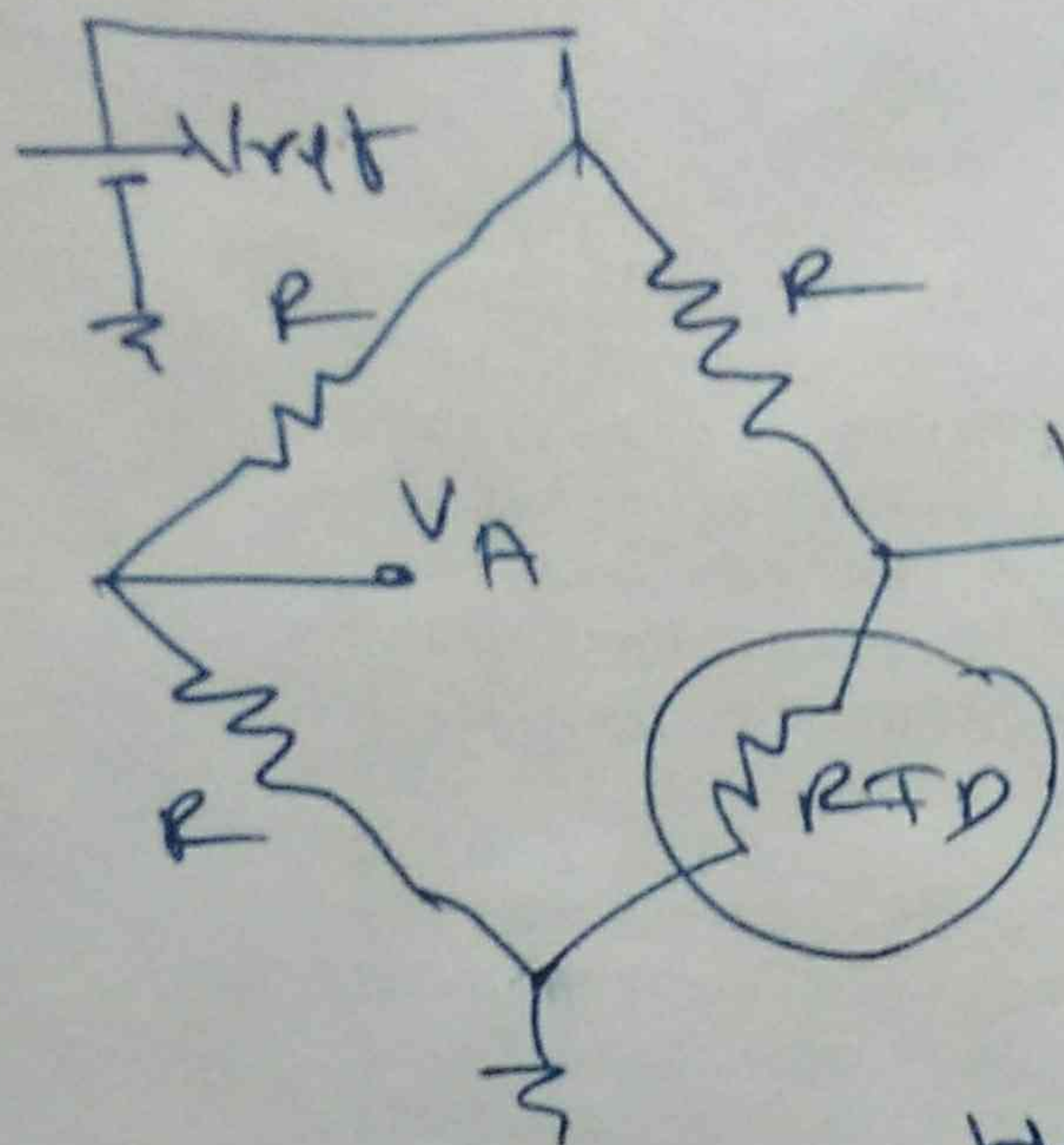
1) what is relevance of v_{cm} , v_1 & v_2

— when Bridge is balanced $V_A = V_B$
($V_A = V_B$)
& o/p will be zero However there

V_B will be common mode voltage $\frac{V_A + V_B}{2} = \frac{V_{ref}}{2}$
— However when Bridge is not balance

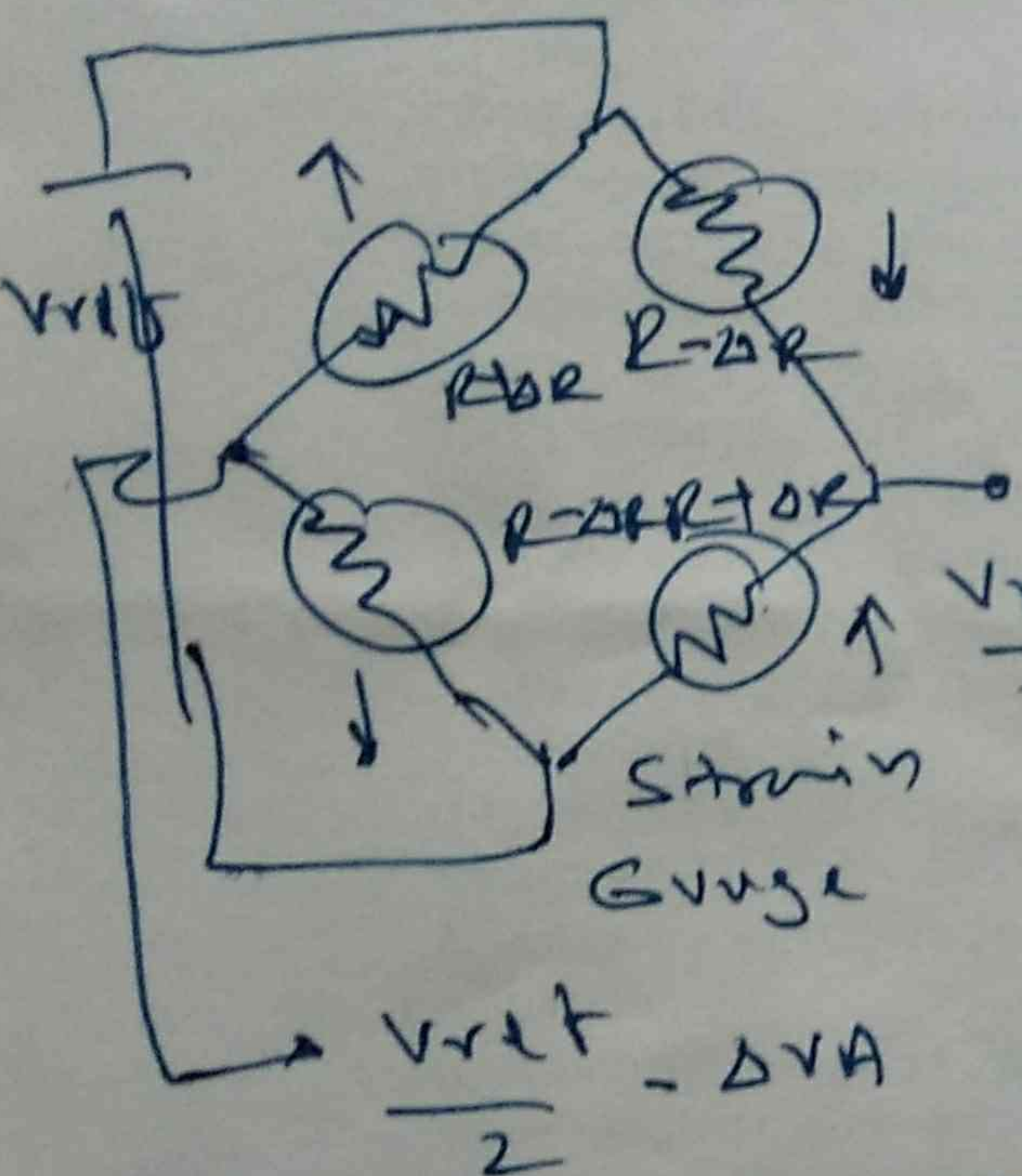
Then $V_A = \frac{V_{ref}}{2}$ and $V_B = \frac{V_{ref}}{2} + \Delta V_B$

Here we will have $v_{cm1} = \frac{\left[\frac{V_{ref}}{2} + \frac{V_{ref} + \Delta V_B}{2} \right]}{2}$
Case 1 $\rightarrow$

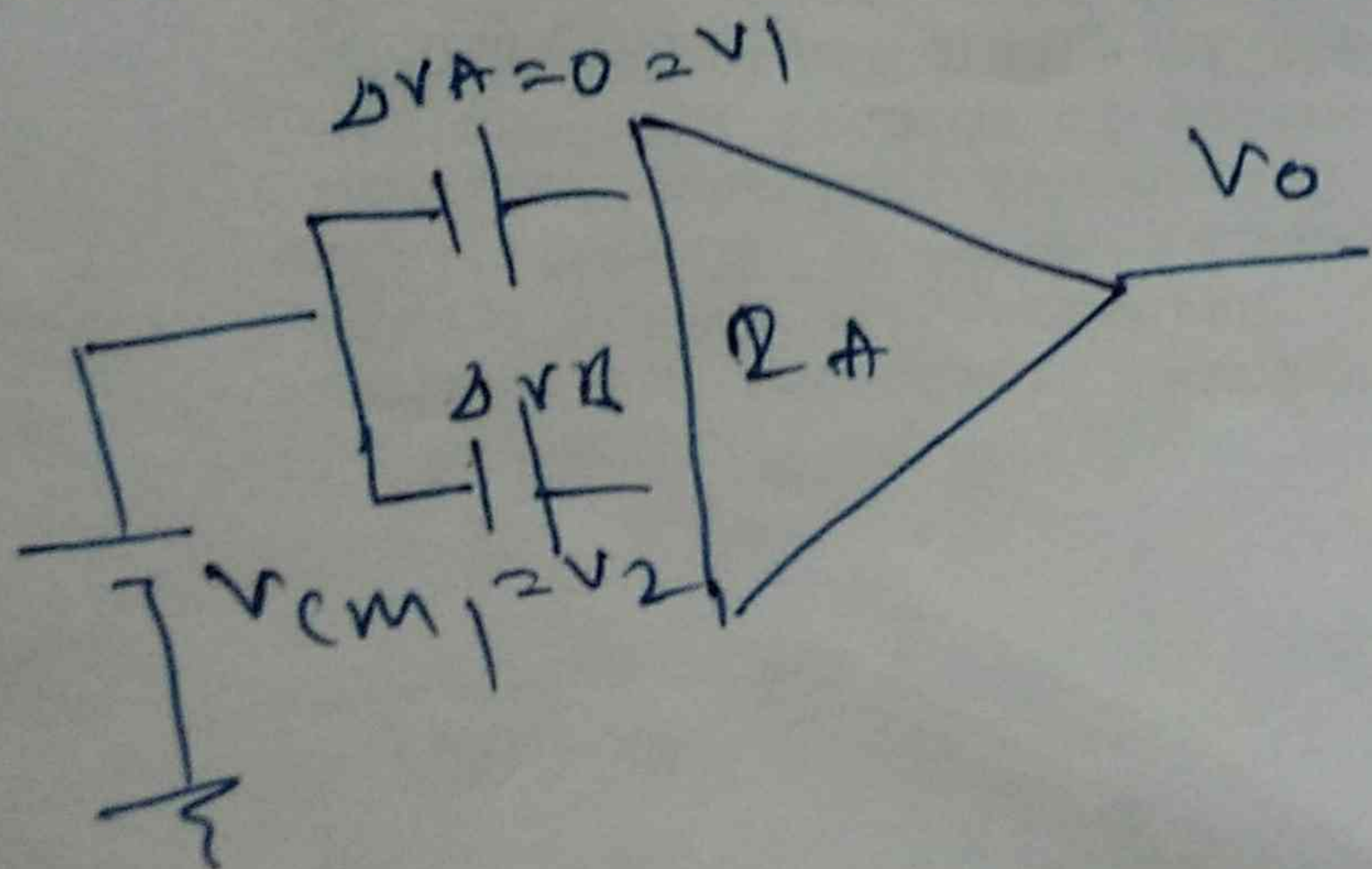


In this case 2

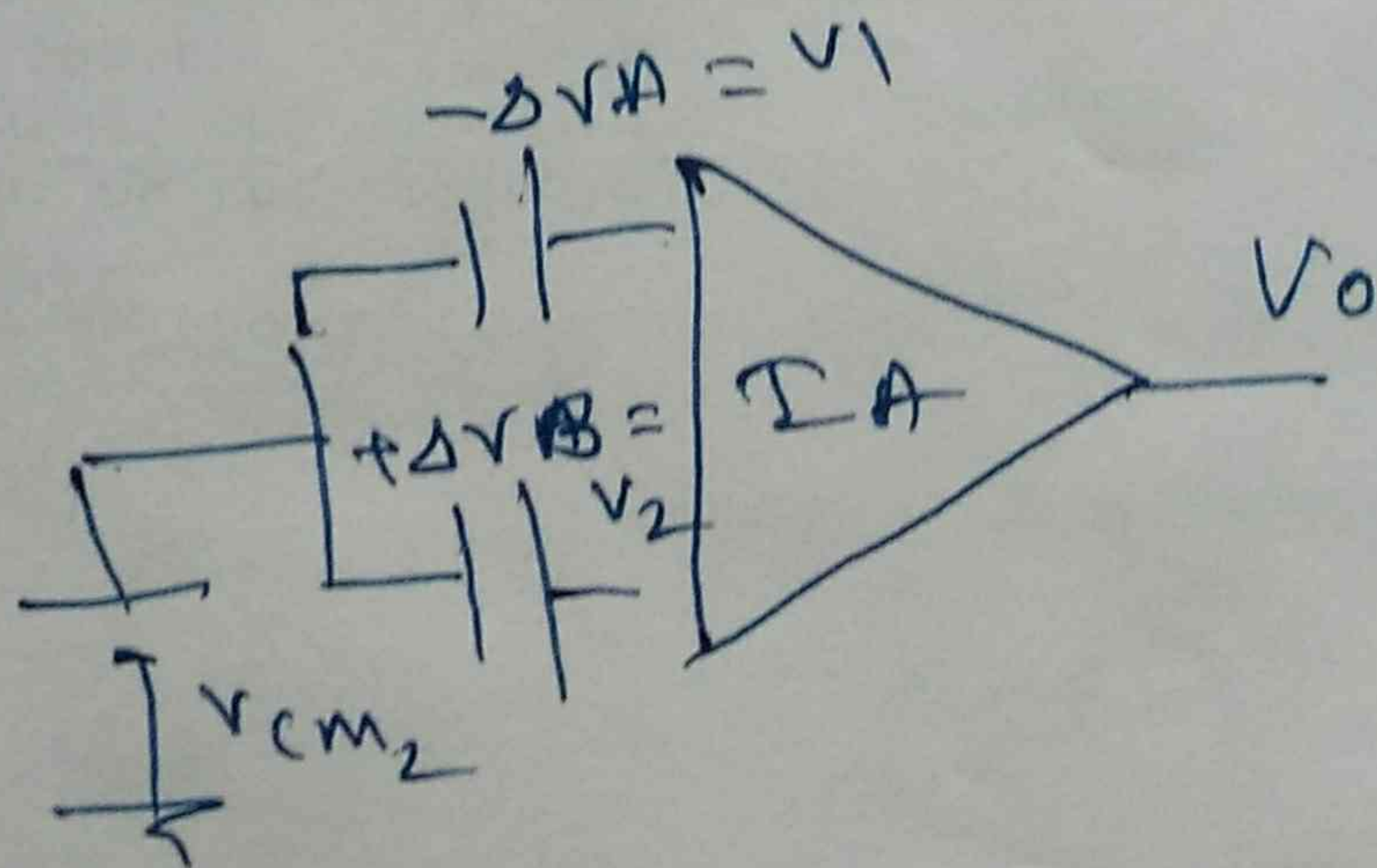
$$v_{cm2} = \frac{\left[\frac{V_{ref}}{2} - \Delta V_A + \frac{V_{ref}}{2} + \Delta V_B \right]}{2}$$



case 1



Case 2



3) Some Discussion

2) Note that $V_{O1} = V_1 + V_{CM} + \frac{R_F}{R_g} V_1 - \frac{R_F}{R_g} V_2$ - (1)

and $-V_{O2} = \frac{R_F}{R_g} V_1 - \frac{R_F}{R_g} V_2 - V_2 - V_{CM}$ - (2)

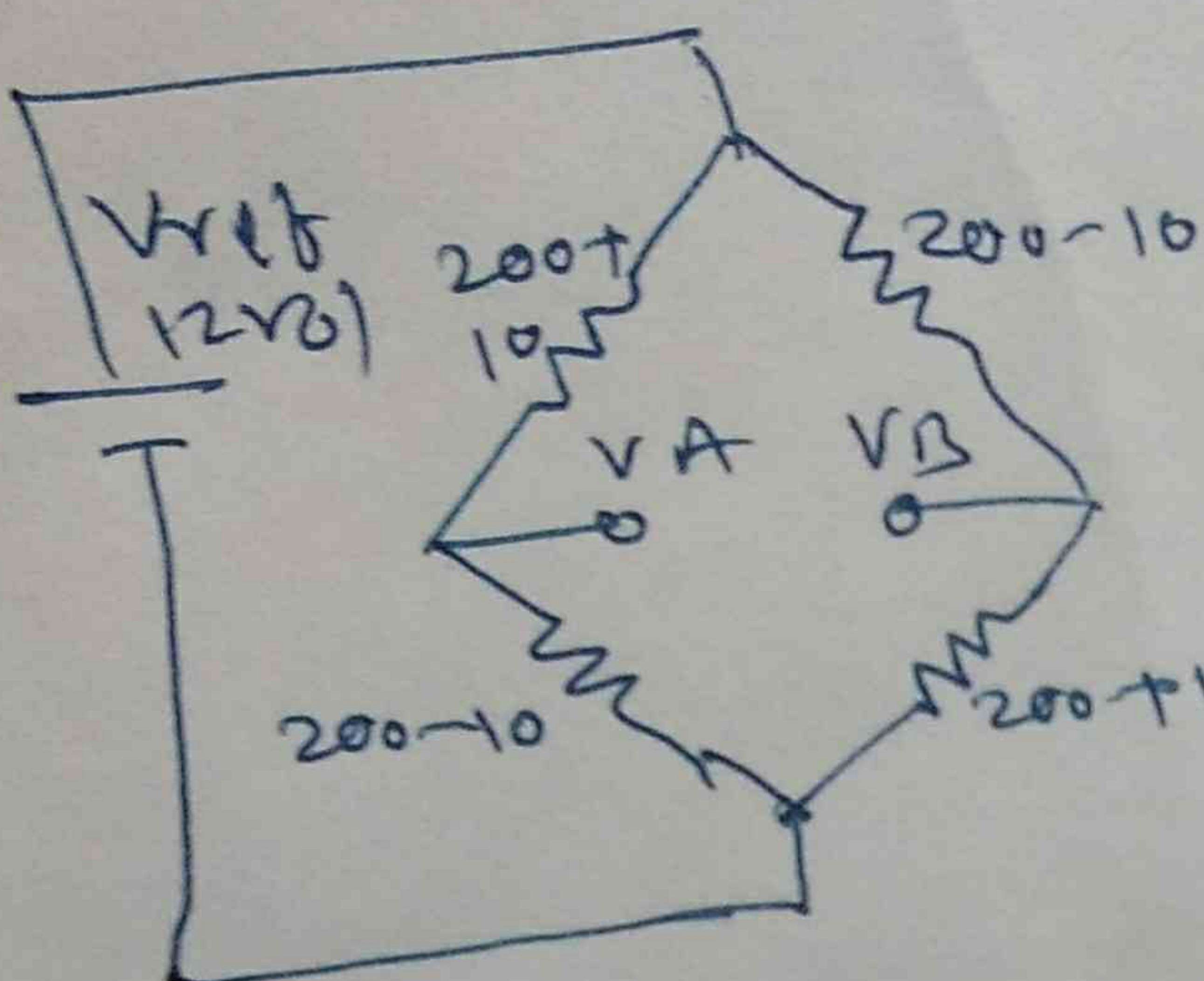
- one can observe that in V_{O1} and V_{O2} V_{CM} is not amplified by $\frac{R_F}{R_g}$ the gain factor.
- the V_{CM} is observed in V_{O1} & V_{O2} which is removed by the subtractor ahead in the design.
- one differential signal $(V_1 - V_2)$ got amplified by $(1 + \frac{2R_F}{R_g})$
- Hence common mode signal got attenuated and differential signal got amplification.

3) This configuration of 3 opamp instrumentation amplifier is extensively used by industry for RTD Based temperature measurement, strain gauge amplifier for designing weighing system and pressure measurements. This amplifier is also used by medical Electronic industry for ECG, EEG signal amplification.

④

Solved Example

- 1) A strain gauge Bridge consists of 4 arms each having resistance of 200Ω at no strain condition. The Bridge is ~~driven from~~ driven by $12V$ Vref
2) under certain strain condition ΔR observed of $\pm 10\Omega$ as shown in the figure



- a) compute common mode signal i) under Bridge Balanced condition ii) under Bridge imbalance condition
b) compute differential mode signal i) under Bridge balanced state ii) Bridge imbalance state
c) compute output voltage of instrumentation amplifier if Bridge ΔR is given to instrumentation Amplifier having following component values
 $R_F = 100K$, $R_g = 50K$, $R_2 = 20K$, $R_1 = 10K$

a-i) when Bridge is balance $V_A = V_B = 6V$
Hence $V_{cm} = \frac{V_A + V_B}{2} = \frac{6+6}{2} = 6V$

a-ii) As Bridge is Balance $V_A = V_B$, $V_D = V_A + V_B = 0$
 $6 - 6 = 0$

b-i) under strain condition $\Delta R = 10\Omega$

Hence $V_A = \frac{190}{200+190} \times 12 = 5.7V$

$V_B = \frac{210}{210+190} \times 12 = 6.3V$

$\therefore V_{cm} = (5.7 + 6.3)/2 = 6V$

$V_d = (5.7 - 6.3) = -0.3V$

c) $V_o = -(V_1 - V_2) \left(1 + \frac{2R_F}{R_g}\right) \left(\frac{R_F}{R_1}\right) = -(-0.3)(5)(2) = 3V$